

DATA CLEANING AND FORMATTING

PROJECT OBJECTIVE:

The main objective of this project is to perform Data cleaning and preparation using Microsoft Excel such as **handling missing values, removing duplicate records, correcting inconsistencies, restructuring columns.**

DATASET DESCRIPTION:

The given Dataset contains information about different products with their Brand and product [ID](#). It also contains information about its price, Quantity and category.

COLUMN DESCRIPTION:

The following are the columns added to the given Dataset.

Column Name	Manufacturing Date	Country Code	Product brand
Description	The Date is splitted from Product ID column	The Country Name is splitted from Product ID column.	Product name and their Brands are merged.

DATA CLEANING PROCESS:

1. Number of Missing Values in price Column: 3

Formula Used: =COUNTBLANK(D2:D35)

2. Missing values can be calculated by using average of the particular column.

Formula Used: IF(ISBLANK(D2), AVERAGE(D2:D35), D2)

3.Products with missing categories:4

For missing values we can take average values.For text values we can replace with null or unknown values.

4.Removing Inconsistent text format:

It can be removed using `PROPER(CLEAN(TRIM(B2)))`

5.Misspellings in category column is replaced using Find and replace function.

Example:Electroni is replaced with Electronics.

6.Removing duplicate values using removing duplicates function.

Number of values removed:3

7.Splitting columns:

Product ID column is split into manufacturing date and country code using text to columns option in data tab.(-) delimiter is used for separation.

8.Merging columns:

Product name and brand name is merged to create brand column.

Formula Used:`CONCATENATE(D2,"-",E2)`

9.Convert price column to currency format:

Change the data type as currency in home tab to change value in price column.

10.Manufacturing date to DD-MM-YYYY Format:

We have no value for year in dataset,the given dataset is formatted by data option in home tab.

11.Conditional formatting in price column:

Color bars are applied to price column based on their price range(small to large)

12.To highlight cells in electronics category using custom rule:

In home tab,select option called highlight cell rules in conditional formatting.select text that contains electronics in category option.

Manufacturing Date	Country Code	Product ID	Product Name	Brand Name	Product Brand	Price	Quantity	Category
28-JAN	US	28-JAN-US	Laptop	Dell	Laptop-Dell	₹ 1,000.00	30	Electronics
15-FEB	US	15-FEB-US	Sneakers	Nike	Sneakers-Nike	₹ 80.00	15	Fashion
3-MAR	US	03-MAR-US	Coffee Maker	Keurig	Coffee Maker-Keu	₹ 130.00	40	Kitchen
11-APR	US	11-APR-US	Smartphone	Samsung	Smartphone-Sams	₹ 900.00	25	Electronics
22-MAY	US	22-MAY-US	Backpack	North Face	Backpack-North F	₹ 70.00	20	Unknown
7-JUN	UK	07-JUN-UK	Headphones	Sony	Headphones-Sony	₹ 309.35	45	Electronics
19-JUL	UK	19-JUL-UK	T-Shirt	Adidas	T-Shirt-Adidas	₹ 30.00	5	Fashion
23-AUG	UK	23-AUG-UK	Blender	Ninja	Blender-Ninja	₹ 90.00	35	Kitchen
5-SEP	UK	05-SEP-UK	Tablet	Apple	Tablet-Apple	₹ 500.00	50	Electronics
14-OCT	UK	14-OCT-UK	Hiking Boots	Timberland	Hiking Boots-Timb	₹ 130.00	10	Outdoor
17-JUN	IN	17-JUN-IN	Laptop	HP	Laptop-HP	₹ 950.00	25	Electronics
25-NOV	AU	25-NOV-AU	Sneakers	Adidas	Sneakers-Adidas	₹ 90.00	40	Unknown
8-DEC	DE	08-DEC-DE	Coffee Maker	Nespresso	Coffee Maker-Nes	₹ 120.00	35	Unknown
18-FEB	CA	18-FEB-CA	Smartwatch	Fitbit	Smartwatch-Fitbit	₹ 150.00	15	Electronics
16-APR	ES	16-APR-ES	Headphones	Bose	Headphones-Bose	₹ 250.00	20	Electronics
21-AUG	CA	21-AUG-CA	Laptop Bag	Samsonite	Laptop Bag-Sams	₹ 50.00	35	Accessories
20-AUG	CN	20-AUG-CN	Smartwatch	Huawei	Smartwatch-Huaw	₹ 160.00	15	Electronics
27-JAN	IT	27-JAN-IT	Laptop	Asus	Laptop-Asus	₹ 980.00	10	Electronics
1-MAR	UK	01-MAR-UK	Sunglasses	Oakley	Sunglasses-Oakley	₹ 150.00	15	Fashion
14-AUG	US	14-AUG-US	Camping Tent	Coleman	Camping Tent-Cok	₹ 309.35	10	Outdoor
14-MAY	RU	14-MAY-RU	Camera	Nikon	Camera-Nikon	₹ 700.00	50	Electronics
9-JAN	CA	09-JAN-CA	Microwave	Panasonic	Microwave-Panasc	₹ 80.00	20	Kitchen
19-JUL	BR	19-JUL-BR	Fitness Tracker	Xiaomi	Fitness Tracker-Xi	₹ 150.00	30	Unknown
29-SEP	CA	29-SEP-CA	Smartphone	Google	Smartphone-Goo	₹ 800.00	45	Electronics
3-JUN	CA	03-JUN-CA	Sunglasses	Ray-Ban	Sunglasses-Ray-B	₹ 309.35	25	Fashion
11-JUL	CA	11-JUL-CA	Blender	Vitamix	Blender-Vitamix	₹ 400.00	40	Kitchen
7-MAR	CA	07-MAR-CA	Dress	Zara	Dress-Zara	₹ 60.00	30	Fashion
13-APR	CA	13-APR-CA	Toaster	Hamilton	Toaster-Hamilton	₹ 40.00	10	Kitchen
24-MAY	CA	24-MAY-CA	Fitness Tracker	Garmin	Fitness Tracker-G	₹ 130.00	5	Electronics
2-DEC	CA	02-DEC-CA	Jeans	Levi's	Jeans-Levi's	₹ 50.00	50	Fashion
9-JUL	FR	09-JUL-FR	Watch	Casio	Watch-Casio	₹ 100.00	20	Accessories

1. Number of missing values in price column: 3
Formula Used: `CountBlank(D2:D35)`
2. Missing values can be replaced with average of the particular column.
Formula Used: `IF(ISBLANK(D2),AVERAGE(D2:D35),D2)`
Average: 309.35
3. products with missing categories: 4
For missing numerical values we can take average but for text values we can not predict it .so we can replace it with unknown or null values.
4. Inconsistent text format in product name can be removed using
Formula: `PROPER(CLEAN(TRIM(B2)))`
5. Misspellings in category column is replaced using find and replace function: For example: Electroni is replaced by Electronics
6. Removing duplicate values using removing duplicates function.
Number of values removed: 3
7. Splitting columns: Product ID column is split into manufacturing date and country code columns.
Function used: Text to columns option in data tab. To separate them (-) delimiter is used.
8. Merging columns: Product name and brand name is merged to create product brand column.
Formula Used: `CONCATENATE(D2,"-",E2)`
9. Convert value in price column to currency format:
In home tab change the data type as Currency for price column.
10. Convert manufacturing date to DD-MM-YYYY format:
Since we have no value for year in given dataset, the given date has been formatted by date option in home tab with date and month are displayed.
11. Conditional Formatting in price column:
Color bars are applied to price column base on the price range (small to large)
12. To highlight cells in electronics category:
In home tab, there is option called highlight cell rules in conditional formatting. In that we should give text that contains electronics in category option.

Summarization:

In this project missing values are verified properly and text formats are standardized. The duplicate records are removed and splitting and merging columns are also performed. Given dataset is formatted and conditional formatting is applied based on the given criteria. The dataset is completely cleaned and formatted and it is transformed with use of excel formulas and functions and result is attached above.